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Source: *The Academy of Management Executive* (1993-2005), Vol. 15, No. 1, Creating Wealth in Organizations (Feb., 2001), pp. 139-148

Published by: [Academy of Management](#)

Stable URL: <http://www.jstor.org/stable/4165716>

Accessed: 20-05-2015 15:23 UTC

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How entrepreneurial firms can benefit from alliances with large partners

Sharon A. Alvarez and Jay B. Barney

Executive Overview

Alliances between entrepreneurial and large firms can create economic value. In many circumstances, however, much of that value is appropriated by the larger partner. In these situations, the performance and even long-term survival of entrepreneurial firms can be put at risk. This article describes the conditions under which the value created by alliances will be appropriated by the large firms, and describes actions that entrepreneurial firms can take to appropriate more of the value created by these alliances.

A study by Andersen Consulting predicted that the value of all the strategic alliances in the world could reach \$40 trillion by the year 2004.¹ A growing number of these alliances exist between large firms and entrepreneurial firms. In the U.S. pharmaceutical industry, for example, alliances between biotechnology firms and pharmaceutical firms increased 341 percent between 1993 and 1995. Similar growth rates can be found in the telecommunications, Internet, electronics, and oil and gas industries.

Large firms in alliances with entrepreneurial firms gain access to new technologies, state-of-the-art engineering talent, and an overall inventive capability. Moreover, by developing alliances with numerous entrepreneurial firms, large firms can partially manage the technological uncertainty they face in their competitive environment. Given these advantages, it is not surprising that large firms often seek out entrepreneurial firms as alliance partners.³

What do the entrepreneurial firms gain from their alliances with large firms? Alliances with a large firm may give an entrepreneurial firm a social legitimacy not otherwise available to it. Large firms often have distribution, manufacturing, marketing, and other organizational resources that an entrepreneurial firm needs to commercialize its technology. Large firms may also have the financial capability needed to bring a technology to market.

Indeed, an alliance between an entrepreneurial and a large firm can instantly improve the economic prospects of an entrepreneurial firm. For example, the stock price of Net2Phone, an entrepreneurial New Jersey-based Internet phone service company, jumped over 50 percent right after it announced strategic alliances with Compaq Computer and Sprint.⁴

However, our research suggests that there are often important differences between the extent to which large and entrepreneurial firms benefit from an alliance. While large firms are usually able to gain access to an entrepreneurial firm's new technology through an alliance, the long-term success of entrepreneurial firms can actually suffer from their alliances with large firms. Indeed, in some circumstances, alliances with large firms can even threaten the survival of an entrepreneurial firm.

While large firms are usually able to gain access to an entrepreneurial firm's new technology through an alliance, the long-term success of entrepreneurial firms can actually suffer from their alliances with large firms.

Over the past three years, we studied 128 alliances between large and entrepreneurial firms in three industries.⁵ In almost 80 percent of these al-

liances, managers from entrepreneurial firms felt unfairly exploited by their large firm partners. Consider, for example, a manager from a company that had developed an important new technology for the biotechnology industry, and entered into an alliance with a large partner in order to commercialize this technology. After several years of working together, the large firm had gained access to this new technology and had successfully commercialized it. The entrepreneurial firm, on the other hand, was bankrupt. A senior manager described his feelings about his large-firm partner in this way:

They are just bullies. They had money, political clout, and staying power. They had a market capitalization of \$13 billion. At that time we had a market cap of maybe \$15 or \$20 million—on a good day. It was very much a David and Goliath story. There was nothing that we could do to stop them. It was cold and calculated on their part. And the real losers are strewn about the roadway here—our employees, the 160 people who lost their jobs between June and September of last year.

Not all entrepreneurial firms have this experience with their large-firm partners. Sometimes entrepreneurial firms are able to gain access to their large-firm partner's skills and use those skills to grow and develop. For example, Network Solutions, Inc., an Internet company with a government contract for registering Internet domains such as .com, .net, and .org, has been very successful in its alliances with Netscape and Yahoo.⁶ This article offers some suggestions for managers in entrepreneurial firms to help them appropriate more of the economic value created by their alliances with large firms.⁷

Entrepreneurial Firms at Risk in Alliances with Large Firms

Our research shows that entrepreneurial firms are most at risk of being taken advantage of by large alliance partners when the only resource they bring to these alliances is a new technology. Two things happen in this setting. First, to the extent that this new technology has market potential, the large firm is often able to realize this potential. Second, entrepreneurial firms in these alliances are often unable to grow and prosper, even if their technology has significant market potential. Thus, while these alliances often create economic value, most of this value is appropriated by the large firm.

Complementary Resources in Alliances

In some industries, entrepreneurial firms most frequently develop important new technologies. In the biotechnology industry, for example, new gene therapies and drug-delivery systems have largely been developed in entrepreneurial firms. In the nearly two-decade fight against AIDS, the first company to reach Phase III trials of an AIDS vaccine was VaxGen, Inc., a small entrepreneurial firm. Three entrepreneurial firms, Valentis, Inc., Vical, Inc., and Vascular Genetics Inc., are developing radical treatments for heart disease. Targeted Genetics Corp. is developing gene therapy to treat cystic fibrosis, while Cell Genesys, Inc., Avigen Inc., and Chiron Corp. are all targeting hemophilia. And the list of technologically inventive entrepreneurial firms in biotechnology goes on and on. Similar lists of inventive entrepreneurial firms exist in the electronics, Internet, and other industries.

There are several reasons why entrepreneurial firms in some industries are often more technologically inventive than large firms. Entrepreneurial firms often can attract the most technically competent scientists and engineers. They do this by compensating employees at least partly through stock and stock options—a form of compensation that promises great wealth if an entrepreneurial firm successfully introduces new technologies to the market. Large firms are often unable to provide such wealth-creating opportunities, and thus are often less able to attract the most technically competent scientists and engineers as employees.⁸

Entrepreneurial firms often can attract the most technically competent scientists and engineers.

Because employees often have a significant ownership stake in entrepreneurial firms, they have very strong incentives to successfully develop new technologies at the lowest cost. Large firms usually do not have these same strong incentives in place, and are often less efficient in developing new technologies.⁹

Finally, because of their size, communication among employees in small entrepreneurial firms is typically informal and flexible. This kind of communication often facilitates technical invention, and is less common in larger firms.¹⁰

Taken together, these attributes of entrepreneurial firms suggest that, on average, entrepreneurial firms are likely to have more inventive capabilities than large firms. These inventive capabilities—

and the promise of the valuable new technologies they hold—often attract large firms to alliances with entrepreneurial firms.

Of course, large firms bring their own resources to alliances with entrepreneurial firms. They have often developed manufacturing, marketing, sales, distribution, financial, and other organizational resources that an entrepreneurial firm needs to commercialize its new technology. Indeed, it is these organizational capabilities that make a large firm an attractive alliance partner for an entrepreneurial firm with a new technology.

Managers in entrepreneurial firms understand the importance of gaining access to these organizational capabilities. A manager at an entrepreneurial startup in the biotechnology industry put it well:

I do believe that you must have a partner in order to be successful. There is no way that we alone could have commercialized our product globally. There is the possibility that in a niche sort of market in the United States we could have commercialized our product alone. But, the prospect of having to do our own manufacturing and raise the money to manufacture and build our own commercial organization for worldwide operations was just overwhelming.

A manager in an entrepreneurial oil and gas firm expressed his firm's need for a large partner even more emphatically:

Our company was essentially broke. We had no assets, no income, and no property. We had \$300 in the bank and \$1 million worth of debt. Vendors were calling and asking, "Where's my money?" But within the first few hours of walking in the door, [our partner] laid a proposal in front of us to do \$5 million. That got our attention. Without our partner, the economic potential of our firm would never have been realized.

When entrepreneurial and large firms control complementary resources, gains from trade are possible and strategic alliances can create economic value.

Learning about New Technologies and Imitating Organizational Resources

If alliances between large firms and entrepreneurial firms can create economic value, why is most of this value often appropriated by the large firm? The answer depends on the difference between the

rate at which large firms learn about the entrepreneurial firm's technology and the rate at which the entrepreneurial firm is able to imitate the organizational resources of the large firm.

Consider first the rate at which large firms typically learn about an entrepreneurial firm's technology. New technologies are often embodied in discrete products or processes. Large firms can study these products and processes to learn about and evaluate an entrepreneurial firm's technologies. Indeed, entrepreneurial firms often facilitate this learning by explaining in detail what their new technology is and how it can be commercialized. In this way, entrepreneurial firms attract large firms to alliances. Moreover, in order for a large firm to manufacture, market, and distribute an entrepreneurial firm's new technology, it usually must have a detailed understanding of how it works. Thus managers in an entrepreneurial firm have strong incentives to communicate as much information as they can to a large firm about a new technology to increase the probability that this technology will actually be commercialized.

A senior manager at a large publicly traded company in the communications network business that has alliance relationships with several smaller entrepreneurial firms understands how easy it usually is for a large firm to learn about an entrepreneurial firm's technology:

The absorption of technology, I've found not to be a problem. In fact, that's the thing that gets done the quickest. Everybody can see clearly how the technology synergies are going to evolve; everybody understands technology synergies in partnerships.

Although it is usually easy for a large firm to learn about an entrepreneurial firm's technology, it is often very difficult for the entrepreneurial firm to learn about and imitate the large firm's organizational resources and capabilities. These capabilities can be very complex. They may have developed over decades and often reflect complex organizational routines taken for granted within a large firm. Moreover, unlike new technologies, which

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are often embodied in discrete products or processes, organizational capabilities are typically diffused throughout a large firm. A large firm's marketing skill can't be taken into a laboratory and studied. Entrepreneurial firms usually do not have the management or financial resources to fully understand, evaluate, and imitate a large firm's organizational capabilities.¹¹

This difference between the rate at which large and entrepreneurial firms learn about each other's resources in an alliance can create significant problems for entrepreneurial firms. These differences can create what has been called a learning race. The first partner in an alliance to learn what it intended to learn from its partner can withdraw from that alliance at low cost. This, of course, cuts short the other firm's ability to learn from the alliance. Since it is usually easier for larger firms to learn about a partner's technology than it is for a smaller firm to learn about its partner's organizational capabilities, entrepreneurial firms usually lose this learning race.¹²

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Actions that Hurt Entrepreneurial Firms' Performance

In general, once a large firm has learned about the new technologies provided by its entrepreneurial partner, it can begin to underinvest in its relationship with the entrepreneurial firm. This can happen in at least two ways. First, it can shift organizational resources away from supporting the alliance to other activities in the large firm. Second, it can place unreasonable demands on the entrepreneurial firm—demands that can hurt its ability to create economic value.¹³

A small entrepreneurial electronics firm with three full-time employees learned about unreasonable demands from its larger-firm partner, which had approximately \$2 billion in sales. According to the president of the entrepreneurial firm:

Probably the best example of how things eventually went in the wrong direction in our alliance is that, as we grew, we were constantly asked to make modifications, to do more research and development work, and to enhance the definition of the scope of our work as it was originally outlined. And we weren't adequately compensated for this ad-

ditional work. This meant that we were constantly short of funds and in a very uncomfortable situation. All along, our partner told us that they were going to put in money. But they never did, knowing full well that we were bleeding to death. Matters got to the point where we litigated against them. When we got to that point, of course, there was no hope that there would ever be anything other than a battle. The cost of this lawsuit exceeded \$4 million.

The large partner began making unreasonable demands after the alliance began to grow, after it had probably learned about the new technology.

Reducing Risks to Entrepreneurial Firms

Our research suggests that the ability of large firms to learn about an entrepreneurial firm's new technology faster than an entrepreneurial partner can imitate a large firm's organizational resources puts the entrepreneurial firm at risk. What can an entrepreneurial firm do to reduce these risks? Table 1 is a summary of five alternatives that entrepreneurial firms can incorporate in their practices.

Going it alone

An entrepreneurial firm may decide not to cooperate with any large firms in commercializing its technology. While this may be an option for some entrepreneurial firms, realizing the full commercial value of a new technology often requires organizational capabilities that most entrepreneurial firms simply do not possess. Moreover, it can be very costly and time-consuming for these firms to try to develop these organizational capabilities on their own. Thus, going it alone is usually not a viable alternative for an entrepreneurial firm. A manager in a small biotechnology firm with less than \$10 million in sales describes the impracticality of going it alone:

In order for us to deliver technology to the market, it may not be feasible for us to go it alone because the health-care market is so immense and we need more commercial power than we can muster. Clearly, the need for partners is the reality for success as an entrepreneurial corporation.

A senior manager at an entrepreneurial software company in the information technology field made similar observations:

Table 1
Entrepreneurial Firm Alternatives in Technology Commercialization

Alternative	Firm Execution	Advantages	Disadvantages
Go it alone.	Acquire and build organizational capabilities internally: administrative, marketing and sales, managerial, technological, manufacturing.	Retains value and benefits of technology R&D and commercialization.	Costly and time consuming. Building capabilities internally has a long time horizon.
Slow the large firm's rate of learning.	Limit the large firm's access to the entrepreneurial firm's technology. Provide the large firm with access to specific technological streams under negotiation and not to the core technology.	Keeps the large firm from appropriating the entrepreneurial firm's technology.	Slows the commercialization of the technology, delaying the entrepreneurial firm's cash flow.
Use detailed and elaborate contracts to define the alliance relationship.	Engage an attorney familiar with alliance agreements to craft the contract. Perform due diligence.	Provides milestone timelines and specific terms and goals.	Contracts cannot fully and unambiguously anticipate all the ways the large firm might adversely affect the entrepreneurial firm. Costly to enforce.
Build a relationship of trust.	Keep lines of communication open with partners. Do not promise to deliver more than what can be delivered.	Enhances the value of the alliance by not having to depend solely on written contracts and reduces worry about the partner appropriating wealth.	Over-reliance on trust can leave the entrepreneurial firm exposed to exploitation by the large firm.
Bring other resources to the alliance besides a single technology.	Maintain the ability to be inventive and produce a stream of new technologies.	Provides strong incentive for the large firm to continue investing in the alliance relationship.	Provides large investments in basic R&D.

We provided some pieces of the solution, but not all, and customers like complete solutions. So the idea here was complementary products—let's align ourselves with best-in-class—so if we don't have a mediation-software product, let's align ourselves with the best-in-class providers in that space. If we don't have a credit-scoring or a fraud solution, let's align ourselves with best-in-class and that's how we came in contact with our alliance partner.

Slowing the large firm's learning rate

An entrepreneurial firm can form an alliance with a large firm, and then try to slow the rate at which its partner learns about its new technologies. If this learning rate can be slowed enough, the large firm may never learn all it wants about the entrepreneurial firm's technology, and may never have an incentive to begin to underinvest in the alliance.

Entrepreneurial firms can limit the large firm's access to its technology to a small number of specific inputs and outputs. It can embed its core technological innovation within less important technological innovations, thereby making it more difficult for the large firm to identify the most important innovations. Or it can simply be slow in responding to the large firm's technical questions. In the extreme, it can even provide misleading answers to these questions.

Entrepreneurial firms can limit the large firm's access to its technology to a small number of specific inputs and outputs.

However, this approach to solving the entrepreneurial firm's problems is itself problematic. After all, the entrepreneurial firm needs the organizational capabilities of the large firm to fully commercialize its new technology. In order to attract a

large firm with the right set of capabilities, the entrepreneurial firm will generally need to describe its technology in some detail. Moreover, once a relationship between a large and entrepreneurial firm is established, the large firm must generally develop a great deal of understanding about an entrepreneurial firm's new technology in order to use its organizational capabilities to commercialize that technology. To the extent that an entrepreneurial firm is successful in slowing the rate at which the large firm learns about the new technology, the entrepreneurial firm is also delaying the eventual commercialization of its technology. This, of course, delays any positive cash flows that product commercialization could create.

Using an alliance contract

An entrepreneurial firm can carefully craft an alliance contract that protects itself from future opportunistic actions of its large-firm partners. One example of this kind of contract is a licensing agreement between an entrepreneurial firm and a large firm that specifies how an entrepreneurial firm's patented technology will be applied by the large firm, and how royalty payments will be allocated.

A great deal of care needs to go into developing a strategic-alliance contract. Large firms often have long due-diligence checklists to evaluate potential alliances. These checklists can then be used to craft a contract that reduces threats in the relationship. Entrepreneurial firms need to invest in their own due diligence and craft alliance agreements that protect themselves from future actions by large-firm partners.¹⁴

While carefully crafted alliance contracts can help reduce the risks of aligning with a large-firm partner, such contracts have important limitations. No matter how elaborate the contract, it cannot fully and unambiguously anticipate all the different ways that a large firm can adversely affect an entrepreneurial firm's performance. Alliances between entrepreneurial and large firms often operate in rapidly changing technological and economic environments. In the face of such uncertainty, it can be very difficult to write alliance contracts that completely protect an entrepreneurial firm's interests. A manager of a biotechnology company with \$25 million in sales describes his experience with contract negotiations:

Sometimes companies want to negotiate the whole damn thing downstream, including the final terms right up front, before they do anything. We tend to discourage that. We try to

get the process going, get into feasibility, and during that time maybe we can come to an agreement on the principal terms, milestones, royalty, and other basic stuff. Sometimes we'll get through two stages, and then we've got to renegotiate something again.

Even if an entrepreneurial firm could write an alliance contract that covered all contingencies, the costs of enforcing these contracts can be substantial. For instance, its large partner's activities

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ultimately led a small electronics firm to sue. But the small firm could not afford the substantial legal bill this action required. Ultimately, a large firm's financial resources often mean that it can hold on while the entrepreneurial firm struggles to enforce what are often unenforceable contracts.

Building trust

Another option an entrepreneurial firm might pursue is building a relationship based on trust with large partners. Our research shows that some entrepreneurial firms believe an open and trusting relationship with a large alliance partner will mitigate the larger firm's propensity to underinvest in the relationship once it has learned what it needed from the alliance.

Much has been written recommending trust as an important component of an alliance relationship.¹⁵ And there is little doubt that trust can enhance the value of alliances when both partners are trustworthy. However, in the context of learning races, an overreliance on trust can put an entrepreneurial firm at significant risk.¹⁶ An executive at a software company with 32 employees clearly understands the limitation of an overreliance on trust:

My biggest concern about technology exchange is that you potentially really hurt your own ability to market. If you've developed something of great value, to make it available to an alliance partner in anticipation of getting something in the future that will make your own firm successful is foolish. All that you've really done as a small company is enable a competitor who can outmarket you

any time that they want. Technologists would undoubtedly tell me I'm way out in left field on this, but it's my sense that all you end up doing is enabling a competitor that can beat the tar out of you in the marketplace.

Even if a large firm enters an alliance in good faith and never has any intention of taking advantage of an entrepreneurial firm, the possibility of underinvestment still exists. While an alliance with a large firm may be an immense undertaking and a significant part of an entrepreneurial firm's business, it is often either insignificant or one of several alliances that the large firm has. Not uncommonly, those who negotiate an alliance for a large firm do not actually manage the alliance. Thus, while an alliance may be entered into in good faith, new alliance managers from the large firm may not be part of the trusting relationship that was formed when the alliance was created. An executive at a small biotechnology firm with fewer than 100 employees understands the implications of new personnel for using trust to manage an alliance well:

People who make the deals and people who live with the deals aren't always the same people. So if you have two thousand employees, the handful of people that put the deal together often don't communicate to the people who are actually going to be carrying the deal forward. People are just sort of thrown into this, they have never been contacted or approached, and their input never obtained. They are just handed this alliance, and that is a problem.

Bringing inventive capabilities to the alliance

An entrepreneurial firm can bring other resources besides a new technology to an alliance. If these other resources are difficult for the large firm to learn about, the latter will not be able to quickly extract all it needs from the entrepreneurial firm, and will not be able to systematically underinvest in the relationship.

Our research shows that the most important of these other resources is inventive capability, that is, the ability to be a stream of new technologies for the large firm into the future, not just a source of a single new technology now. If an entrepreneurial firm offers not just its current new technology, but the promise of additional new technologies in the future, then its large partner will have strong incentives to continue investing in the relationship. The interest, and thus the relationship, will con-

tinue as long as the large firm still has valuable things to learn. An alliance executive with an entrepreneurial drug-delivery company, with approximately \$25 million in sales and a history of successful alliances, understands this principle very well:

It's not an alliance, it's a series of alliances. It's a series of technologies that put proteins into a sustained-release format. That can be done with many proteins. We have two publicly announced agreements. One was with a large firm for human-growth hormone. And we have a number of others that we're working on, they just haven't been disclosed yet. And there's a subsidiary technology as well that we've just taken over that expands that capability into small molecules, and we just announced two days ago an agreement with a large firm for an unnamed compound.

With such a string of new technologies and product opportunities, this firm has been highly successful in alliances with larger firms.

There are several ways that entrepreneurial firms can bring the promise of a string of new technologies to alliances. For example, rather than investing in a single technology, entrepreneurial firms can invest in a bundle of related new technologies simultaneously. To extract value from strategic alliances, entrepreneurial firms should be somewhat diversified in their technology development efforts.¹⁷

To extract value from strategic alliances, entrepreneurial firms should be somewhat diversified in their technology development efforts.

Rather than invest only in minor modifications of established technologies, an entrepreneurial firm may want to also invest in more basic R&D. Not only is basic R&D usually more difficult for large firms to understand, it often has broader technological implications than R&D on minor technological improvements. These broader technological implications may hold the promise of a string of technologies.

Firms that invest in platform technologies that can be used as a basis for developing numerous other technologies are more likely to bring the possibility of a string of technologies to the alliance than firms that invest in a stand-alone new

Table 2
How Entrepreneurial Firms and Large Firms Can Benefit from Alliances

Firm type	Manage technology carefully	Recognize the different rates of learning between firms	Understand the need that each firm is trying to fill through the alliance	Reduce risk
Entrepreneurial firm	Bring a string of technologies to the alliance or have the potential to generate a string of technologies.	Slow the large firm's rate of learning about the technology. Do not overexpose firm's technological capabilities early in the partnership.	Large firms often need the inventiveness of entrepreneurial firms. Once the large firm has the new technology, it can usually exploit the technology.	Perform due diligence on the large firm under consideration. Be cautious to prevent excessive, perhaps debilitating, appropriation of the alliance benefits by the large firm.
Large established firm	Choose entrepreneurial partners capable of generating several technology streams.	Select entrepreneurial-firm partners with sufficient management skills to learn large-firm organizational capabilities.	Does the entrepreneurial firm want to remain independent or be acquired? Does it want to remain the same size or grow?	Form alliances with entrepreneurial firms that have managers capable of understanding what is required to make the alliance successful.

technology. This kind of investment can also reduce the risk of being taken advantage of.

Regardless of how an entrepreneurial firm's R&D is organized, the critical task is to demonstrate to potential partners that the entrepreneurial firm possesses inventive capabilities. On average, large firms will have more difficulty developing these capabilities than entrepreneurial firms. If an alliance holds the promise of gaining access to these valuable and costly-to-imitate capabilities, a large firm has very strong incentives to continue to invest in its alliance. The threat of losing the learning race is then reduced for the entrepreneurial firm, and opportunities are enhanced to appropriate any value that the alliance with the large firm creates.

Put differently, firms that pursue a diversified technology-development strategy, that invest in more basic R&D, or that invest in platform technologies, are less likely to be exploited in alliances with large firms. Alliances with large firms are particularly attractive to such firms. Firms that can bring only a single new technology to an alliance will generally find these alliances to be unsatis-

factory. These firms should pursue alternative approaches for gaining access to a partner's organizational capabilities. For example, these firms may want to ally with other smaller firms, either in traditional alliances or in networks.

Building Inventive Capability in Entrepreneurial Firms

Alliances with large organizations are clearly not organizational panaceas for entrepreneurial firms. They hold enormous potential for firms that bring inventive capability and the promise of several new technologies to an alliance. However, for entrepreneurial firms with only a single technology, alliances with large firms promise as much danger as opportunity. Such firms may see their technologies commercialized through these alliances, but will often not be able to share in the wealth created by these commercialization efforts.

Similarly, entrepreneurial firms that form alliances with trustworthy partners, exercise due diligence in choosing those partners, and carefully craft strategic-alliance contracts, can protect themselves in strategic alliances with large firms. But these activities often cannot compensate for an entrepreneurial firm is bringing only a single new technology to an alliance. Without the promise of a string of new potentially valuable technologies,

Firms that can bring only a single new technology to an alliance will generally find these alliances to be unsatisfactory.

large firms have few incentives to maintain alliances with entrepreneurial firms, especially once the large firm has learned about the single new technology. And once the large firm starts withdrawing from the alliance, it is difficult for the entrepreneurial firm to protect itself.

These observations suggest that technology does not enable entrepreneurial firms to create value and appropriate wealth in alliances with large firms. Most technology can be imitated at a lower cost than the initial innovation, and thus is not a source of sustained competitive advantage. Rather, the inventive capability within entrepreneurial firms—a capability that large firms usually value but cannot develop or imitate—makes it possible for entrepreneurial firms to create value and appropriate wealth through alliances with large firms. Table 2 summarizes the findings presented in this article.

An alliance manager at an information-technology firm summarizes much of what we learned about these alliances:

Alliances, in my definition, are broad and long lasting. An alliance is a broad-based agreement that we will collaborate over a very broad set of issues. So it is more than just one project, more than just one program, more than just one technology where we have similar interest areas, where we are engaged in similar areas of research, and have lots to exchange. These common interests can be a product, products, or areas of intellectual growth. It could be the development of a process or an improved understanding of a basic scientific principle. Whatever the source, you have developmental interests together that result in products and processes with market potential.

Acknowledgments

The authors would like to thank Dale Meyer for his advice and encouragement; the special editors, in particular Duane Ireland; the two blind reviewers; and David Deeds for his thoughtful comments. This research was supported entirely by the Kauffman Foundation.

Endnotes

¹ See CMA Management. Alliance management. December/January 2000:73.

² The growth of strategic alliances between large and entrepreneurial firms has been documented in: In praise of business alliances. *Business Week*. 25 October 1999, 206. In 1989, Barley, Freeman, and Hybels documented 3,441 ties distributed among 10 different types of relationships, including equity holdings, joint ventures, R&D agreements, product-development agree-

ments, supply agreements, manufacturing agreements, marketing agreements, and licensing agreements. Barley, S. R., Freeman, J., & Hybels, R. C. 1992. Strategic alliances in commercial biotechnology. In N. Nohria & R. G. Eccles, (Eds.), *Networks and organizations*. Cambridge, MA: Harvard Business School Press. Fisher notes that alliances between biotechnology firms and pharmaceutical firms numbered 85 in 1993 and had increased to 375 in 1995, for a 341 percent increase. Fisher, L. 1996. How strategic alliances work in biotech? *Technology*, Winter. Hitt, M. A., Tyler, B. B., Hardee, C., & Park, D. 1995. Understanding strategic intent in the global marketplace. *The Academy of Management Executive*, 9:12–19.

³ The use of alliances to manage technological uncertainty was first discussed by Kogut, B. 1991. Joint ventures and the option to expand and acquire. *Management Science*, 37:19–33.

⁴ Frankel, M. Closing bell. *Business Week*. 6 September 1999, 42.

⁵ The study spanned a three-year period, from June 1996 through July 1999, during which 40 entrepreneurial firm alliance executives were interviewed, followed by two phases of mail surveys. In this research, an entrepreneurial firm was defined as a firm with fewer than 150 employees that had been in existence for less than five years. Entrepreneurial firms in our sample averaged \$25 million in sales, while their large-firm partners averaged \$1 billion in sales. A first set of 25 interviews was started in 1996. These interviews were used to develop a mail survey, which was then sent to all publicly traded entrepreneurial biotechnology firms in 1996. Two additional industries were added to the biotechnology sample, with 15 new interviews conducted in 1999. These two industries were the information-technology industry and the oil-and-gas industry. Only quotations from the second part of the study are used in this article. All executives interviewed were promised confidentiality. Quotes from these interviews have been edited for the ease of the reader. A final mail survey was sent to 657 firms, and a total of 128 usable surveys was returned.

⁶ Indeed, these alliances have been largely responsible for a rise in Network Solutions' stock from \$59 to \$165 per share. See Clausen, J. Network Solutions soars, despite the coming loss of its exclusive accord for Internet domain names. *New York Times*. 22 December 1998, 5. In December 1992, Network Solutions entered into a cooperative agreement with the National Science Foundation. Network Solutions provided Internet domain-name registration services for .com, .net, .org, .edu, and .gov. The Department of Commerce took over the administration of the cooperative agreement in 1998. The agreement was later extended to September 30, 2000, when Network Solutions entered a series of agreements with the Department of Commerce. On March 7, 2000, Network Solutions merged with VeriSign, Inc. VeriSign issued 1.075 shares of stock for every one share of Network Solutions stock (Network Solutions, 1999 Annual Report). VeriSign stock was at \$84.88 a share on Jan. 16, 2001.

⁷ In this article, the economic value created by an alliance is equal to the difference between the present value of the firms working together in an alliance and the sum of the present value of each of these firms operating independently. This economic value becomes economic wealth when it is appropriated by the owners and other stakeholders of each of these firms.

⁸ This phenomenon was described by Zenger, T. R. 1989. Organizational diseconomies of scale: Pooling vs separating labor contracts in Silicon Valley (California). Unpublished doctoral dissertation. University of California, Los Angeles.

⁹ See Williamson, O. E. 1975. *Markets and hierarchies: Analysis and antitrust implications*. New York: The Free Press; and Williamson, O. E. 1985. *The economic institutions of capitalism*. New York: The Free Press.

¹⁰ See Cooper, A. 1964. R&D is more efficient in small companies. *Harvard Business Review*, May/June: 75-83.

¹¹ These observations about a large firm's organizational capabilities draw on the resource-based view of the firm. See Barney, J. B. 1995. Looking inside for competitive advantage. *The Academy of Management Executive*, 9:49-61.

¹² Learning races in the context of international strategic alliances were first described in Hamel, G., 1991. Competition for competence and inter-partner learning within international strategic alliances. *Strategic Management Journal*, 12:83-103.

¹³ These are examples of moral hazard and hold-up, respectively. See Barney, J. B. & Ouchi, W. G. 1986. *Organizational economics*. San Francisco: Jossey-Bass.

¹⁴ Large firms often have checklists with over 400 items. See Hubbard, G., Lofstrom, S., & Tully, R. 1994. Diligence checklists: Do they get the best answers? *Mergers & Acquisitions*, 29:33-38. Most entrepreneurial firms are unlikely to have such extensive due-diligence checklists. Still, they can usually do more due diligence than they do. In our research, it was not unusual for

an entrepreneurial firm to spend not even half a day on due diligence. For more information on this point see Cyr, D. 1999. High tech, high impact: Creating Canada's competitive advantage through technology alliances. *The Academy of Management Executive*, 13(2):17-28.

¹⁵ See for example Barney, J. B., & Hansen, M. H. 1994. Trustworthiness as a source of competitive advantage. *Strategic Management Journal*, 15:175-190.

¹⁶ The limitations of trust in these settings is described in more detail in Deeds, D. L. & Hill, C. W. L. 1998. An examination of opportunistic action within research alliances: Evidence from the biotechnology industry. *Journal of Business Venturing*, 14: 141-163.

¹⁷ Of course, to pursue this diversification strategy, an entrepreneurial firm must have the required technological and managerial resources. If the cost of acquiring or developing these resources is less than the value at risk from large-firm partners' underinvesting in alliances with these entrepreneurial firms, this diversification strategy may be justified.



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